NYA-IOI-001 / VER-IOI-001 — Inferred Ownership Intelligence Layer

Operating Design Document and Implementation Specification — v1.2 (Synthesis-Integrated)

> **Status:** Approved future scope and committed. Sequenced after RRI, after the cross-document scanning contract is stable enough to emit governed bundle entities and relationships, **and after NYA-INTEL-SYNTH-001 (Deterministic Intelligence Synthesis) has shipped its first slice**. This is not a deferral or scope removal. v1.2 is a complete document and supersedes v1.1 in full; no reader needs v1.1.

Field	Value
Version	v1.2 — synthesis-integrated revision (complete, self-contained)
Supersedes	v1.1 architecture revision (23 July 2026); v1.0
Date	13 October 2026
Architecture owner	Nyansapo IOI bounded context
Synthesis role	**Upstream evidence authority.** IOI consumes the signed synthesis Assessment for a subject before any of its own acquisition. Synthesis never consumes IOI output.
Ceres role	Evidence grounding provider only; no inferred-edge ownership
Veredis role	Trigger policy, orchestration, consumption, rendering and review submission
Decision posture	Review-only; `CONTEXT_ONLY` maximum decision effect; composing IOI with synthesis does not raise the cap
Current dependencies	Cross-document scanning (in development); NYA-INTEL-SYNTH-001 first slice (specified v0.5.0, awaiting founder rulings)

Operating phrase: *AOT proves. Synthesis establishes. RRI connects. IOI infers. Ceres grounds.*

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1. Executive summary

IOI — Inferred Ownership Intelligence — is a future, review-only intelligence engine that identifies *possible* ownership, beneficial-control, proxy, nominee, successor, shadow-director, executive-control and operating-control relationships where confirmed AOT evidence is incomplete, unavailable, opaque or contradicted, **and where deterministic synthesis has not already established or disproved the proposition from admissible structured sources.**

IOI is a first-class bounded context inside Nyansapo, adjacent to AOT, Synthesis, RRI and Ceres. It is not implemented inside Ceres, not inside Synthesis, and not as a Veredis inference engine.

> **Core rule:** IOI may identify what deserves investigation. It may not decide what is legally true, write confirmed ownership, alter a synthesis assertion, or alter a deterministic Veredis decision.

1.1 Product promise

Where ownership cannot be confirmed by AOT and cannot be established by Synthesis, Veredis can present reviewable inferred ownership/control signals with source lineage, contradiction handling, temporal role history, cross-document context, the synthesis basis they rest on, and explicit separation from both confirmed AOT and established Synthesis.

1.2 Layer definitions

Layer	Meaning	Primary function	Maximum authority
AOT	Automated Ownership Traversal	Confirmed ownership/control traversal using admitted structured evidence	Deterministic only where governed policy permits

Layer	Meaning	Primary function	Maximum authority
Synthesis	Deterministic Intelligence Synthesis (NYA-INTEL-SYNTH-001)	Establishes, from admissible structured facts under governed rules, what is established / unresolved / disproved / conflicted for a question about a subject; emits signed Assessments	`CONTEXT_ONLY`; deterministic; no probability
RRI	Relationship Risk Intelligence	Broader relationship and adverse-context graph	Review, context and EDD
IOI	Inferred Ownership Intelligence	Possible ownership/control/proxy signals where proof is missing and synthesis is silent or unresolved	Review-only; `CONTEXT_ONLY`
Ceres	Legal and policy grounding	Retrieves grounded legal/context passages and source status	Context provider; no relationship decision authority

2. Revision history

2.1 v1.1 → v1.2 (this revision)

Area	v1.1	v1.2
Synthesis	Not mentioned (predates NYA-INTEL-SYNTH-001).	Synthesis is the **first input** to every IOI evaluation (§10). Established facts enter IOI as S1 evidence; hypotheses/indicators target IOI acquisition; closed/conflicted propositions constrain IOI signals.
Vocabulary	IOI used `INFERRED_*`, `ASSOCIATION_ONLY`, "inference", "evidence", "contradiction" with no separation from synthesis terms.	**Vocabulary separation is enforced on the IOI side only** (§5.5): synthesis terms are reserved; IOI renames `ASSOCIATION_ONLY` → `ASSOCIATION_SIGNAL`; consumed synthesis assertions carry `SYNTH_*` labels; the copy guard forbids cross-use. Synthesis vocabulary is unchanged.
Placement diagram	Four siblings (AOT, RRI, IOI, Ceres).	Five siblings; Synthesis added (§4).
Logical flow	Started at source discovery.	Step 0: obtain the synthesis Assessment (§12.2).
Data model	No synthesis linkage.	`synthesis_basis` on `ioi_inferred_edges`; `SYNTHESIS_ASSESSMENT_SOURCE` family; `ioi_synthesis_consumption` table; `ioi_ingest_candidates` (§16).

Area	v1.1	v1.2
Reverse flow	IOI → AOT candidates only.	Adds IOI → ingest candidates (`POSSIBLE_SYNTHESIS_EVIDENCE`); never IOI → synthesis facts (§14).
Sequencing	After RRI and cross-doc.	Additionally after the synthesis first slice; dependency listed (§24).
Decisions	None open.	§27 lists the v1.2 founder decisions.

2.2 v1.0 → v1.1 (retained)

System placement corrected: IOI is an independent Nyansapo bounded context, not a Ceres submodule and not Veredis-owned. Ceres returns grounded passages and source/legal status only. Veredis owns trigger policy, orchestration, rendering and review submission only. Authoritative review history and all inferred edges remain in Nyansapo. IOI waits for a stable governed cross-document output contract.

3. Current-state grounding

3.1 Nyansapo standing

- **Data substrate:** Nyansapo owns source acquisition, source lineage, canonical entity resolution, provenance, graph contracts, medallion/admissibility and AOT.
- **Confirmed traversal:** AOT remains the only confirmed ownership traversal path.
- **Deterministic establishment:** Synthesis (NYA-INTEL-SYNTH-001) is the only engine that may say a relationship proposition is *established* or *disproved* from structured evidence. Its first slice answers `Q-ASSOC-1` (source-established relationships around a subject) and emits signed, snapshot-pinned Assessments with terminal statuses per rule and binding.
- **Relationship substrate:** RRI represents broader relationships and adverse context; IOI may consume RRI features but retains separate edge types and review state.
- **Ceres:** context-only; supplies grounded legal or policy passages.
- **IOI:** new bounded context with its own schema, APIs, guardrails, audit trace and lifecycle.

3.2 Veredis standing

- Consumes governed Nyansapo outputs; does not write ownership or relationship truth back.
- Owns scan orchestration and deterministic user-facing outcomes; IOI cannot alter them.
- Web, PDF and structured API/VZP output must preserve identical IOI caveats, evidence summaries **and synthesis-basis labels**.
- Captures user actions but sends them to Nyansapo as review events.

3.3 Synthesis standing

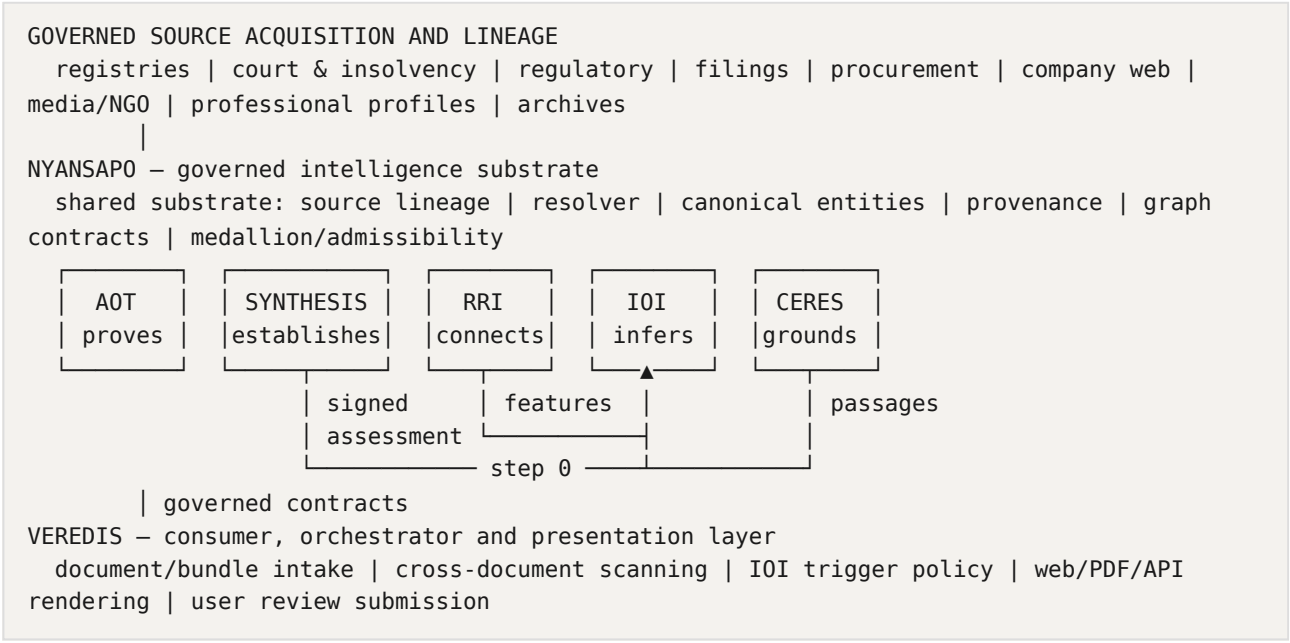
Synthesis is specified (v0.5.0) and awaiting founder rulings FD-IS1–16. Its outputs relevant to IOI: Assessment id, execution header (snapshot, rule-set digest, `as_of`, pinned releases), assertions with class (`FACT`, `DERIVED_FACT`, `INFERENCE`, `CONFLICT`, `HYPOTHESIS`, `INDICATOR`, `GAP`, `UNKNOWN`), per-binding terminal statuses (`EMITTED`, `HYPOTHESIS`, `UNRESOLVED`, `CLOSED`, `SUPPRESSED`), and the explain payload per assertion. Synthesis exposes `/v1/intel/assess/*` under scope `INTEL_READ`. IOI's accommodation requests toward the synthesis programme are listed separately (dev-team message accompanying this document) and do not alter synthesis vocabulary.

3.4 Cross-document scanning standing

Cross-document scanning is in active development. IOI binds only to the stable governed contract produced after cross-document entity reconciliation and role assignment.

> **Sequencing:** IOI is approved and committed. Its synthesis adapter is sequenced after the synthesis first slice; its cross-document adapter after the cross-document contract is stable. IOI core schema, source lineage and evidence contracts may be designed independently.

4. Architecture decision and system placement



4.1 Why IOI belongs in Nyansapo

- IOI depends on canonical identity, source lineage, evidence hashing, resolver state, graph contracts, time-aware relationship history **and signed synthesis Assessments** — all substrate assets.
- Keeping confirmed (AOT), established (Synthesis) and inferred (IOI) relationships physically and contractually separate while resolving all three against the same Golden Records prevents a split-brain graph.
- Authoritative review history, contradiction state and confidence decay must survive across Veredis scans and future consumers.

4.2 Why IOI must not live inside Synthesis

- Synthesis is deterministic: closed expression evaluator, no probability class, admissible structured sources only, no generative or ranking models in its production path, and a constitutional prohibition on promoting association to control (its C2).
- IOI is banded-confidence over S1–S5 sources including media, NGO reports and self-asserted profiles, uses an extractor and a ranker with model versions in its audit trace, and exists precisely to surface *possible* control where proof is absent.
- Merging them would either admit weak sources and probability into synthesis — destroying its contract — or strip IOI of confidence and weak sources, at which point it is no longer IOI.

4.3 Why IOI must not live inside Ceres

Ceres is a retrieval and grounding subsystem, not a probabilistic graph-state owner. Ceres may classify a passage's legal status; it must not emit `INFERRED_CONTROLLER = true`.

4.4 Why Veredis must not own the engine

Veredis would duplicate identity resolution, lineage, graph persistence and evidence decay; inference would sit next to deterministic decision rendering; IOI would become product-specific.

5. Design doctrine, vocabulary separation and non-goals

5.1 Allowed language (IOI outputs)

```
Sources suggest a possible ownership or control relationship between X and Y.
This has not been confirmed through AOT and has not been established by Synthesis.
Review required before reliance.
```

5.2 Forbidden language (IOI outputs)

```
X owns Y.
X controls Y.
Hidden owner found.
Beneficial owner confirmed.
Blocked because of IOI.
Synthesis suggests…            (synthesis does not suggest; it establishes or does not)
Established by IOI.            (IOI never establishes)
IOI inference confirmed.
```

5.3 Non-negotiable constraints

Constraint	Rule
AOT isolation	IOI edges cannot enter confirmed AOT traversal tables or be traversed as confirmed ownership.
Synthesis isolation	IOI output never becomes a synthesis fact, precondition, binding or override. IOI never writes to `intel_*` tables.
Decision cap	`IOI_MAX_DECISION_EFFECT = CONTEXT_ONLY`. No BLOCK, CLEAR, ESCALATE or HOLD. Composition with synthesis (also `CONTEXT_ONLY`) does not raise the cap.
Evidence floor	No signal without at least one evidence item, source family, retrieval time and hash.
Synthesis precedence	A synthesis `CLOSED(FALSE)` on the same proposition and binding forces any IOI signal asserting it to band `CONTRADICTED`. A synthesis `EMITTED(INFERENCE)` on the same proposition means IOI emits **no competing signal**; it references the assessment instead.

Constraint	Rule
Contradiction-aware	Material contradictions (including synthesis `CONFLICT`) must be retained, scored and shown.
User review is not truth	A user confirmation records usefulness or case judgment; it creates neither confirmed ownership nor an established synthesis fact.
No source laundering	Several weak sources do not become a primary source by aggregation; a synthesis Assessment is one S1 evidence item per cited assertion, not many.
Temporal honesty	Role history and control claims are time-bounded and decay or become stale. Synthesis-derived evidence carries the Assessment's `as_of` and snapshot, not IOI's retrieval time.
Cross-doc restraint	Bundle co-occurrence triggers investigation; it is not ownership evidence.

5.4 Non-goals

- Replacing AOT or lowering AOT admissibility standards.
- Re-implementing, extending or relabelling Synthesis; changing any synthesis vocabulary.
- Turning Ceres into a general inference or graph engine.
- Building a generic adverse-media product under the IOI name.
- Treating professional profiles, media or user confirmations as ownership proof.
- Re-implementing cross-document scanning inside IOI.
- Hidden decision effects or unreviewable model conclusions.

5.5 Vocabulary separation (enforced in IOI only)

Synthesis is a filed programme; its vocabulary is **reserved** and unchanged. IOI adjusts on its side so that no reviewer, renderer or API consumer can confuse a deterministic synthesis result with an IOI signal.

Concept	Synthesis term (reserved — IOI must not use for its own outputs)	IOI term (v1.2)
A rule-derived conclusion from admissible structured facts	`INFERENCE` (class), `EMITTED` (status)	— IOI never produces one. When consumed: `SYNTH_ESTABLISHED`
A possible relationship from weighted sources	—	`INFERRED_*` signal types (unchanged names; "inferred" in IOI means *possible, banded, review-only*)
Association without control	`ASSOCIATION_CLUSTER_SUPPORTED`	`ASSOCIATION_SIGNAL` (renamed from `ASSOCIATION_ONLY`)

Concept	Synthesis term (reserved — IOI must not use for its own outputs)	IOI term (v1.2)
Missing-evidence state	`HYPOTHESIS` + `INDICATOR`, `UNRESOLVED`, `GAP`	When consumed: `SYNTH_HYPOTHESIS`, `SYNTH_UNRESOLVED`, `SYNTH_GAP`. IOI's own: `INSUFFICIENT_EVIDENCE` (unchanged)
Disproof	`CLOSED(FALSE)`	When consumed: `SYNTH_CLOSED`. IOI's own: `CONTRADICTED_SIGNAL`
Contradiction between admissible facts	`CONFLICT`	When consumed: `SYNTH_CONFLICT`. IOI's own: contradiction types in §17.3
Evidence unit	fact (`fact_key`), assertion (`assertion_key`)	`evidence item` (`ioi_evidence_items.id`)
Confidence	none (no probability class)	`confidence_band` (`LOW/MEDIUM/HIGH/CONTRADICTED/STALE/INSUFFICIENT`) — never applied to `SYNTH_*` items
Admissibility	`ADMISSIBLE / ADMISSIBLE_ADVISORY / INADMISSIBLE_*`	`source_reliability_band` S1–S5 and acquisition status (§9.2)
"Established"	reserved for synthesis `EMITTED` assertions	IOI copy guard rejects "establish" for any IOI signal
"Suggests / possible"	never used by synthesis	reserved for IOI signals; copy guard rejects it for `SYNTH_*` items

Copy guard rules (build-time and request-time, see §20.6): (a) the tokens `INFERENCE`, `EMITTED`, `ESTABLISHED` may appear in IOI output only inside a `SYNTH_*` label or a verbatim synthesis assertion; (b) the words *suggests*, *possible*, *may* may not modify a `SYNTH_ESTABLISHED` item; (c) `confidence_band` must be null on every `SYNTH_*` evidence item; (d) no IOI signal type name may collide with a synthesis predicate name (CI check against the synthesis predicate enum).

6. Responsibility and ownership matrix

Capability	Nyansapo IOI	Synthesis	Ceres	Veredis
Source acquisition and family policy	Owns (IOI families)	Owns (admissible structured families)	Selected legal-source support	No
Canonical entity resolution	Consumes shared resolver; owns IOI linkage	Consumes with strict floor	No	Supplies scan/bundle candidates
Deterministic establishment of a proposition	**No**	**Owns**	No	No
Grounded legal passages	Consumes	Never (C6)	Owns	No

Capability	Nyansapo IOI	Synthesis	Ceres	Veredis
Inferred-edge generation	**Owns**	No	No	No
Confidence, corroboration, contradiction (banded)	Owns	No (terminal statuses only)	May supply passage status	No
Temporal role lineage	Owns	Owns tenure model for its facts	No	No
Authoritative IOI graph and review history	Owns	No	No	Submits review events
Synthesis Assessment consumption	Consumes (step 0)	Produces	No	May request both
Ingest candidates from IOI findings	Files	Receives via governed source admission (not into facts)	No	No
Cross-document reconciliation	Consumes contract	No	No	Owns scan-side feature
When to request IOI	Exposes capability	No	No	Owns orchestration policy
Web/PDF/API presentation	Projects payload	Projects Assessment	No	Owns
Deterministic Veredis decision	No	No	No	Owns; neither IOI nor Synthesis can alter

7. Source family registry

Every source connector is registered by family, reliability class, access method, allowed use, retention policy and extraction contract. The family identifies the evidential nature of the record; the source name identifies the specific registry, court, outlet, platform or provider.

7.1 Official, legal and primary-source families

Source family	Examples	IOI use	Default treatment
SYNTHESIS_ASSESSMENT_SOURCE (new)	Signed NYA-INTEL-SYNTH-001 Assessments	Established facts, derived facts, closed propositions, conflicts, hypotheses with indicators	**S1**; `confidence_band` null; carries synthesis `as_of`, snapshot and signature; one evidence item per cited assertion

Source family	Examples	IOI use	Default treatment
CORPORATE_REGISTRY_SOURCE	National company registries, officer and beneficial-owner filings	Directors, officers, PSCs, charges, historical filings	High; structured ownership may become AOT candidate; where synthesis already consumes the same registry release, IOI cites the synthesis fact rather than re-reading the row
LEI_RELATIONSHIP_SOURCE	GLEIF Level 2 and related LEI records	Parent, child, entity lineage	High for declared relationships
SECURITIES_FILING_SOURCE	Annual reports, major holdings, related-party and issuer filings	Ownership, subsidiaries, officers, related parties	High
COURT_RECORD_SOURCE	Judgments, pleadings, asset recovery, civil and criminal proceedings	Alter ego, nominee, shadow control, ownership allegations or findings	High but legal-status aware
BANKRUPTCY_INSOLVENCY_SOURCE	Petitions, insolvency reports, administrator statements	Hidden interests, creditor influence, related-party transfers	High if official
REGULATORY_ENFORCEMENT_SOURCE	Regulator findings, sanctions notices, asset freezes	Evasion networks, control findings, restricted-party links	High
PROCUREMENT_SOURCE	Awards, supplier records, contracting notices	Repeated related-party suppliers, operating networks	Medium/high
ASSET_REGISTRY_SOURCE	Land, vessel, aircraft, mining, licence records	Asset control and beneficial-interest indicators	Medium/high

7.2 Open intelligence and contextual families

Source family	Examples	IOI use	Default treatment
COMPANY_WEB_SOURCE	Leadership, board, founder, team pages	Executive role lineage, self-published operating roles	Medium/high
MEDIA_SOURCE	Investigative media, reputable adverse media, trade press	Control allegations, successor entities, historic lineage	Medium/high if corroborated
NGO_REPORT_SOURCE	Anti-corruption, sanctions-evasion, kleptocracy reports	Network and control allegations	Medium/high
ARCHIVED_WEB_SOURCE	Historical company pages, removed biographies	Historical role continuity, successor signals	Medium; staleness-aware
BUSINESS_DIRECTORY_SOURCE	Industry and chamber directories	Weak role or association evidence	Low/medium

Source family	Examples	IOI use	Default treatment
PROFESSIONAL_PROFILE_SOURCE	Executive profiles, business-network platforms	Self-asserted roles, founder claims, board history, affiliations	Medium but capped
SEARCH_DISCOVERY_SOURCE	Search results, index metadata	Discovery only until the underlying source is retrieved	Not evidence

7.3 Reliability bands

Band	Meaning	Examples	Effect
S1	Primary official/legal, or signed synthesis Assessment	Registry filing, judgment, regulator decision, issuer filing, synthesis assertion	Strong IOI support; registry rows may create separately validated AOT candidates; synthesis items are never re-banded
S2	Official self-published	Company site, annual report, official biography	Strong role signal; source-specific caveat
S3	Reputable secondary	Investigative media, trade press, NGO report	Corroborated and contradiction-aware
S4	Self-asserted professional	Professional profile or executive network	Confidence-capped; role signal, not ownership proof
S5	Discovery or weak	Search snippet, directory, mirror	Discovery or low confidence only

8. Professional-profile and LinkedIn source treatment

Professional-profile data is valuable for executive role chronology, founder identity, board participation, employment overlap and company affiliation. It is not inherently ownership evidence. IOI treats it as a source family with explicit acquisition lineage and confidence caps.

8.1 Source types

```
SELF_ASSERTED_FOUNDER
SELF_ASSERTED_OWNER
SELF_ASSERTED_EXECUTIVE
SELF_ASSERTED_BOARD_MEMBER
SELF_ASSERTED_COMPANY_AFFILIATION
EXECUTIVE_ROLE_LINEAGE
FOUNDER_SUCCESSOR_SIGNAL
```

8.2 Access lineage classes

Access class	Meaning	Treatment
OFFICIAL_API_APPROVED	Platform API access approved for the use and storage pattern	Ingest only within granted limits
LICENSED_PROVIDER	Contract permits downstream professional/company data use	Ingest under licence and lineage reference
USER_SUPPLIED_EXPORT	Customer or subject supplies an export or evidence	Ingest with consent/attestation reference
MANUAL_REVIEW_CAPTURE	Analyst records a limited case-level role fact	Case-level evidence; not bulk assumption
GOVERNED_WEB_ACQUISITION	Collection method approved under source policy for the specific source	Ingest with access method and policy decision recorded
SEARCH_DISCOVERY_ONLY	Search result identifies a potential profile or company page	Discovery only
PROHIBITED_OR_UNAPPROVED	Method lacks source-policy approval	Reject or quarantine

8.3 Confidence caps

Claim	IOI classification	Maximum without corroboration
CEO / Managing Director / President	SELF_ASSERTED_EXECUTIVE	LOW or ASSOCIATION_SIGNAL
Founder / Co-founder	SELF_ASSERTED_FOUNDER	MEDIUM; not proof of current ownership
Owner / Principal / Managing Partner	SELF_ASSERTED_OWNER or POSSIBLE_CONTROL	MEDIUM; review-only
Board member / Chair	GOVERNANCE_INFLUENCE_SIGNAL	MEDIUM
Advisor / Consultant	ASSOCIATION_SIGNAL	LOW
Connections, likes, comments or endorsements	Out of scope	Reject

A self-asserted role that synthesis has already **established** from a registry appointment (`SYNTH_ESTABLISHED` on the same person/company/role) is recorded as *corroborated by synthesis* and does not need its own cap; a self-asserted role that synthesis has **closed** (e.g. the registry shows the appointment resigned before the claimed period) is `CONTRADICTED`.

9. Source acquisition and lineage contract

The acquisition method is part of the evidence chain. An IOI evidence item is invalid if the system cannot state where it came from, how it was accessed, when it was retrieved, what claim was extracted and which versions resolved the entities.

9.1 Required lineage fields

```
source_family
source_name
source_record_id              # for SYNTHESIS_ASSESSMENT_SOURCE: "
<assessment_id>/<assertion_key>"
source_url_hash
source_url_redacted
access_method                 # for synthesis items: "INTERNAL_SERVICE:/v1/intel/assess"
access_authority_or_policy_ref
source_reliability_band
retrieval_timestamp
publication_date              # for synthesis items: the Assessment's as_of
evidence_hash                 # for synthesis items: the Assessment's canonical_sha256
claim_text_redacted
language
translated_claim_text_redacted
extraction_method             # for synthesis items: "SYNTH_ASSERTION_PASSTHROUGH"
extractor_version
resolver_version              # for synthesis items: the binding is inherited, not re-
resolved
supports_or_contradicts
allowed_use_classification
retention_expires_at
synth_label                   # SYNTH_ESTABLISHED | SYNTH_HYPOTHESIS | SYNTH_UNRESOLVED |
SYNTH_GAP | SYNTH_CLOSED | SYNTH_CONFLICT | null
```

9.2 Acquisition statuses

Status	Meaning	Pipeline effect
ADMITTED	Source and acquisition method satisfy policy and parsing requirements	Eligible for scoring
REVIEW_REQUIRED	Source or access method needs analyst/policy review	Quarantine from automated inference
DISCOVERY_ONLY	Reference exists but underlying evidence is not retrieved	May launch retrieval; no signal support
REJECTED	Source is unreliable, unverifiable, prohibited or corrupt	Excluded
STALE	Evidence is expired or superseded	Confidence capped and surfaced
SYNTH_SUPERSEDED	The synthesis Assessment consumed has been superseded by a later execution for the subject	IOI signal flagged for re-evaluation; see §10.6

10. Synthesis consumption contract (new)

Synthesis is the first input to every IOI evaluation. The flow is strictly one-way for facts and one narrow channel for ingest candidates.

10.1 Step 0 — obtain the Assessment

Before any source discovery, IOI requests the current synthesis Assessment for the subject (`POST /v1/intel/assess` , question `Q-ASSOC-1` in the first slice; further question types as synthesis adds them). IOI is an internal Nyansapo service and calls synthesis through the same scoped boundary any trusted consumer uses (`INTEL_READ` , internal-service audience, object-level ownership of the assessments it creates). IOI records the consumption in `ioi_synthesis_consumption` (§16) with `assessment_id` , `execution_id` , `snapshot_id` , `rule_set_sha256` , `as_of` .

If synthesis is unavailable, IOI proceeds in **degraded mode**: every signal carries `synthesis_basis = null` and `synthesis_status = UNAVAILABLE` , rendering states that established-fact grounding was not possible, and no `SYNTH_*` constraints are applied. IOI does not fabricate a synthesis result.

10.2 Established facts become S1 evidence

Each synthesis assertion of class `FACT` , `DERIVED_FACT` or `INFERENCE` (status `EMITTED`) becomes one `ioi_evidence_items` row: family `SYNTHESIS_ASSESSMENT_SOURCE` , band S1, `synth_label = SYNTH_ESTABLISHED` , `evidence_hash = canonical_sha256` , `publication_date = as_of` , `confidence_band = null` . The binding (`ch_person_id` , company numbers, tenures) is inherited verbatim; IOI's resolver does not re-resolve identities that arrive this way.

Where an IOI candidate relationship is **the same proposition** as a `SYNTH_ESTABLISHED` assertion on the same binding, IOI emits **no competing signal**. The rendered output references the synthesis assertion (`see established fact asm-…/assertion_key`) and IOI's own signal, if any, is limited to what synthesis did not cover (e.g. a *control* inference layered on a synthesis-established *association*).

10.3 Hypotheses and indicators target acquisition

Each synthesis `HYPOTHESIS` with its `INDICATOR` s becomes (a) an evidence item `SYNTH_HYPOTHESIS` and (b) one or more **targeted acquisition jobs** in `ioi_acquisition_jobs` with `triggered_by {assessment_id, assertion_key, precondition_id, indicator_text, source_families}` . IOI's weaker-source acquisition is thereby aimed at closing a specific named gap. The resulting IOI signal, if any, must state which indicator it addresses and that synthesis could not establish the proposition because that evidence was missing from admissible structured sources.

10.4 Closed propositions and conflicts constrain signals

- A `CLOSED(FALSE)` terminal status on a binding becomes a `SYNTH_CLOSED` evidence item with `supports_or_contradicts = CONTRADICTS` . Any IOI signal asserting the same proposition on the same binding **must** carry `confidence_band = CONTRADICTED` , with the synthesis assertion listed first among contradicting items. IOI never suppresses the signal silently and never overrides the synthesis disproof.
- A synthesis `CONFLICT` assertion becomes a `SYNTH_CONFLICT` evidence item and a source-contradiction entry (§17.3). IOI does not attempt to resolve it.
- `UNRESOLVED` , `GAP` and `UNKNOWN` become `SYNTH_UNRESOLVED` / `SYNTH_GAP` items; IOI proceeds normally and records that synthesis was silent on the proposition.

10.5 Reverse channel — ingest candidates only

IOI output never becomes a synthesis fact, precondition, binding or override (constraint table §5.3). The sole reverse channel: when IOI admits an evidence item from a **structured** source family that synthesis does not yet ingest (e.g. a court register, an insolvency administrator's report), IOI files `ioi_ingest_candidates` row `{candidate_type: POSSIBLE_SYNTHESIS_EVIDENCE, source_family, source_name, evidence_item_id, rationale}`. This is routed to governed source admission as a **request to consider adding a source family** to the synthesis loader. It is never a fact and never bypasses synthesis admissibility.

10.6 Pinning and supersession

Every IOI signal carries `synthesis_basis {assessment_id, execution_id, snapshot_id, rule_set_sha256, as_of}` or null. IOI periodically checks whether a newer synthesis execution exists for the subject; when one does, affected signals are marked `SYNTH_SUPERSEDED` and queued for re-evaluation against the newer Assessment. Until synthesis provides drift (NYA-INTEL-SYNTH-004), IOI performs this check itself by re-requesting an Assessment and comparing `execution_id`.

10.7 Decision-effect composition

Synthesis Assessments and IOI signals are both `CONTEXT_ONLY`. Presenting them together in Veredis does not raise the cap. Veredis must not compute any combined score from a `SYNTH_ESTABLISHED` item and an IOI `confidence_band`.

11. Cross-document scanning integration

11.1 Boundary

Cross-document scanning is upstream of IOI for transaction-bundle cases. Veredis reconciles document parties and discrepancies, then sends the governed bundle context to Nyansapo IOI. IOI performs source-backed investigation against the resolved entities; it does not reparse or reconcile the bundle. For bundle-triggered evaluations IOI still performs step 0 (§10.1) for each resolved `golden_id`.

11.2 Governed input contract

```
bundle_id
documents[]
reconciled_entities[]
party_role_assignments[]
cross_doc_relationships[]
bundle_discrepancy_findings[]
entity_resolution_candidates[]
source_document_pointers[]
audit_trace_id
```

11.3 Trigger patterns

Cross-doc pattern	IOI investigation question	Allowed result
Supplier differs from payment beneficiary	Is the payee an affiliate, proxy, nominee or controlled entity?	INFERRED_PAYMENT_PROXY only with external evidence

Cross-doc pattern	IOI investigation question	Allowed result
Shipper differs from seller	Is the shipper part of the same operating/control network?	INFERRED_TRADE_PROXY or ASSOCIATION_SIGNAL
Consignee differs from buyer	Is the consignee a related party or hidden intermediary?	POSSIBLE_CONTROL or ASSOCIATION_SIGNAL
Repeated unknown intermediary	Is the intermediary a recurring facilitator or front?	INFERRED_PROXY_ENTITY if corroborated
Executive name appears in certificate or declaration	Is the person linked to one or more bundle entities?	EXECUTIVE_ROLE_LINEAGE (or reference to SYNTH_ESTABLISHED appointment)
Shared bank, address or contact details	Common control or service-provider overlap?	ASSOCIATION_SIGNAL unless corroborated; if synthesis established co-location in release R, reference it

> **Hard gate:** Transaction-bundle co-occurrence is not ownership evidence. A cross-document finding may trigger IOI retrieval, but an inferred edge requires independent admitted evidence.

12. Nyansapo IOI bounded context

12.1 Component layout

```
nyansapo/ioi/
  source_registry/        # family definitions, reliability and policy
  synthesis_adapter/      # step-0 request, SYNTH_* mapping, supersession check, ingest-
candidate filing
  acquisition/            # governed adapters and retrieval jobs (incl. indicator-targeted
jobs)
  evidence/               # evidence items and provenance
  resolution/             # Golden Record and person/entity linking (never re-resolves
SYNTH_* bindings)
  candidate_generation/   # possible relationship candidates
  temporal_lineage/       # role history, start/end and succession
  scoring/                # evidence weighting and confidence bands (SYNTH_* items
excluded from banding)
  contradiction/          # conflicts, supersession and decay (incl. SYNTH_CLOSED /
SYNTH_CONFLICT)
  graph/                  # isolated inferred edges
  review/                 # review events and pattern feedback
  projections/            # Veredis/RRI-safe response contracts
  api/                    # entity, bundle, review and admin endpoints
  audit/                  # immutable trace payloads
  guards/                 # AOT isolation, synthesis isolation, decision caps, copy guard
```

12.2 Logical flow

```
golden_id or governed bundle context
  -> STEP 0: synthesis Assessment request and consumption (§10.1–10.4)
  -> source discovery and source-policy validation (indicator-targeted first)
  -> governed acquisition
  -> evidence extraction and legal-status annotation
  -> entity/person resolution (SYNTH_* bindings inherited)
  -> relationship candidate generation (excluding propositions SYNTH_ESTABLISHED on the
same binding)
  -> temporal alignment
  -> corroboration and contradiction analysis (SYNTH_CLOSED forces CONTRADICTED)
  -> confidence banding (SYNTH_* items unbanded)
  -> isolated IOI edge and audit trace (with synthesis_basis)
  -> ingest-candidate filing where applicable (§10.5)
  -> Veredis-safe projection
```

12.3 Persistence boundary

- IOI owns inferred edges and evidence links in isolated tables or schema.
- IOI never writes into confirmed ownership tables or into synthesis `intel_*` tables.
- IOI references canonical Golden IDs and synthesis `assessment_id` / `assertion_key`; it creates no second identity universe.
- Bundle-only unresolved entities may be referenced temporarily until resolver status changes.
- Authoritative review events are stored in Nyansapo and projected to Veredis.

13. Ceres evidence-grounding contract

Ceres is a provider to IOI, not the owner of IOI conclusions. It retrieves relevant legal, regulatory, policy or court passages and returns provenance, passage location, source status and legal posture.

13.1 Ceres may return

```
        {
          "document_id": "court_record_123",
          "source_family": "COURT_RECORD_SOURCE",
          "passages": [
            {
              "text_redacted": "The respondent exercised de facto control ...",
              "source_location": "paragraph 47",
              "legal_status": "ALLEGATION | FINDING | JUDGMENT | GUIDANCE",
              "entities": ["Person A", "Company B"],
              "supports_or_contradicts": "UNASSESSED"
            }
          ]
        }
```

13.2 Ceres must not return

```
INFERRED_CONTROLLER = true
CONFIRMED_OWNER = true
DECISION_EFFECT = BLOCK
PROMOTE_TO_AOT = true
SYNTH_ESTABLISHED = true        # only a signed synthesis Assessment can carry this label
```

13.3 IOI use of Ceres output

IOI resolves the named entities against canonical records; determines whether a passage supports, contradicts or merely mentions a relationship; combines the passage with non-Ceres evidence families; preserves Ceres source location and legal status in the evidence trail. A Ceres passage is never a synthesis fact and is never relabelled `SYNTH_*`.

14. Relationship to AOT, Synthesis and RRI

14.1 AOT boundary

AOT remains the confirmed ownership traversal engine. IOI may create a separately typed investigation candidate when evidence appears capable of validation through official structured sources. The candidate is a work item, not a promoted edge.

```
        {
          "candidate_type": "POSSIBLE_AOT_EVIDENCE",
          "origin": "IOI",
          "status": "NEEDS_STRUCTURED_VALIDATION",
          "allowed_decision_effect": "NONE"
        }
```

14.2 Synthesis boundary

Direction	Allowed data flow
SYNTH → IOI	Signed Assessment: established facts (S1 evidence), hypotheses + indicators (targeted acquisition), closed propositions and conflicts (constraints), unresolved/gap (silence). Explain payloads for rendering lineage.
IOI → SYNTH	**Ingest candidates only** (`POSSIBLE_SYNTHESIS_EVIDENCE`), routed through governed source admission. Never facts, preconditions, bindings or overrides.

14.3 RRI boundary

RRI and IOI are adjacent, not hierarchical. RRI may provide graph features such as common addresses, officers, co-defendants, enforcement associations or shared service providers. IOI determines whether those features, together with source evidence and the synthesis basis, support an ownership/control-specific inference. RRI may later render IOI as a typed sublayer, but IOI retains its own state and caveats.

Direction	Allowed data flow
RRI → IOI	Relationship candidates, graph features, adverse context and source pointers
IOI → RRI	Typed review-only inferred-control edges for optional rendering
IOI → AOT	Investigation candidate only; never a direct confirmed edge
AOT → IOI	Confirmed path status, gaps, unknown upstream nodes and superseding evidence

15. Signal taxonomy and confidence model

15.1 Signal types

```
INFERRED_BENEFICIAL_OWNER
INFERRED_CONTROLLER
INFERRED_NOMINEE_OWNER
INFERRED_PROXY_ENTITY
INFERRED_FAMILY_CONTROL
INFERRED_SHADOW_DIRECTOR
INFERRED_FINANCIER_CONTROL
INFERRED_ASSET_CONTROL
INFERRED_SUCCESSOR_ENTITY
INFERRED_EXECUTIVE_CONTROL
INFERRED_TRADE_PROXY
INFERRED_PAYMENT_PROXY
EXECUTIVE_ROLE_LINEAGE
ASSOCIATION_SIGNAL              # renamed from ASSOCIATION_ONLY (§5.5)
INSUFFICIENT_EVIDENCE
CONTRADICTED_SIGNAL
```

CI check: no signal type name may equal any synthesis predicate (`SHARED_OFFICER_ESTABLISHED`, `REGISTERED_ADDRESS_COLOCATED_IN`, `ASSOCIATION_CLUSTER_SUPPORTED`, `RELATIONSHIP_EVIDENCE_CONFLICTED`, `OWNERSHIP_PATH_CITED`, `SANCTIONS_LISTING_CITED`, and any later addition read from the synthesis rule-set manifest).

15.2 Confidence bands

Band	Meaning	Rendering rule
LOW	Weak, single-source or limited-corroboration signal	Display only when materially relevant; ask for more evidence
MEDIUM	Credible signal with some corroboration	Display source basis, dates and contradictions
HIGH	Strong multi-source signal but still unconfirmed and not established by synthesis	Remain review-only; never render as proof

Band	Meaning	Rendering rule
CONTRADICTED	Material contradictory evidence exists (including any `SYNTH_CLOSED` on the same binding)	Show contradiction prominently; cap reliance
STALE	Evidence is outdated or superseded	Label and reduce confidence
INSUFFICIENT	Mention exists but does not support inference	No active edge; retain rejected candidate only if useful

`SYNTH_*` evidence items never carry a band. A signal resting **only** on `SYNTH_ESTABLISHED` items is not an IOI signal at all — it is a reference to synthesis and is rendered as such.

15.3 Evidence dimensions

Dimension	Purpose
source_reliability	Weight primary, official, self-published, secondary and self-asserted sources appropriately; S1 synthesis items are fixed-weight
independence	Prevent copied or syndicated claims counting as independent corroboration
entity_resolution_confidence	Ensure the claim refers to the correct person/company; inherited for `SYNTH_*` bindings
relationship_extraction_confidence	Whether text actually states or implies the typed relation
legal_status	Allegation, finding, judgment, guidance, self-assertion
recency and temporal fit	Align evidence dates with the claimed control period; synthesis items use the Assessment `as_of`
corroboration and contradiction	Count supporting and conflicting independent evidence; `SYNTH_CLOSED` is a mandatory contradiction
source-access status	Evidence admitted under source policy
synthesis basis	Whether and how the signal rests on, addresses, or is constrained by a synthesis Assessment
review history	Record user/admin outcomes without turning them into legal truth

16. Data model

16.1 Core tables

Table	Purpose
ioi_inferred_edges	Authoritative isolated inferred relationship edge (with `synthesis_basis`)
ioi_evidence_items	Source evidence, claims, hashes, access lineage, legal status, `synth_label`

Table	Purpose
ioi_edge_evidence_links	Many-to-many with support/contradiction role and weight
ioi_temporal_roles	Person/entity role history, start/end dates and source support
ioi_review_events	Immutable user/admin review actions
ioi_pattern_feedback	Aggregate feedback for ranking or suppression policy
ioi_bundle_context_links	Links signal to governed cross-document trigger and document pointers
ioi_aot_candidates	Separate work queue for possible structured AOT validation
ioi_synthesis_consumption (new)	One row per step-0 consumption: subject, assessment_id, execution_id, snapshot_id, rule_set_sha256, as_of, consumed_at, superseded_by
ioi_acquisition_jobs (new)	Acquisition work items incl. `triggered_by` (assessment/assertion/precondition) for indicator-targeted jobs
ioi_ingest_candidates (new)	`POSSIBLE_SYNTHESIS_EVIDENCE` requests to governed source admission

16.2 ioi_inferred_edges

```
id UUID PRIMARY KEY
from_golden_id TEXT NULL
to_golden_id TEXT NULL
from_bundle_entity_id TEXT NULL
to_bundle_entity_id TEXT NULL
inferred_relationship_type TEXT NOT NULL
confidence_band TEXT NOT NULL
internal_score NUMERIC NULL
status TEXT NOT NULL DEFAULT 'ACTIVE'
decision_effect TEXT NOT NULL DEFAULT 'CONTEXT_ONLY'
source_count INTEGER NOT NULL DEFAULT 0
corroboration_count INTEGER NOT NULL DEFAULT 0
contradiction_count INTEGER NOT NULL DEFAULT 0
earliest_evidence_date DATE NULL
latest_evidence_date DATE NULL
effective_from DATE NULL
effective_to DATE NULL
review_status TEXT NOT NULL DEFAULT 'PENDING'
synthesis_basis JSONB NULL          -- {assessment_id, execution_id, snapshot_id,
rule_set_sha256, as_of, addresses_indicator: {assertion_key, precondition_id} | null,
constrained_by: [assertion_key...]}
synthesis_status TEXT NOT NULL      -- CONSUMED | UNAVAILABLE | SUPERSEDED
created_at TIMESTAMP NOT NULL
updated_at TIMESTAMP NOT NULL
CHECK (decision_effect = 'CONTEXT_ONLY')
```

16.3 ioi_evidence_items

```
id UUID PRIMARY KEY
source_family TEXT NOT NULL
source_name TEXT NOT NULL
source_record_id TEXT NULL
source_url_hash TEXT NULL
source_url_redacted TEXT NULL
access_method TEXT NOT NULL
access_authority_or_policy_ref TEXT NULL
source_reliability_band TEXT NOT NULL
publication_date DATE NULL
retrieval_date TIMESTAMP NOT NULL
evidence_hash TEXT NOT NULL
excerpt_redacted TEXT NULL
language TEXT NULL
translated_excerpt_redacted TEXT NULL
extraction_method TEXT NOT NULL
extractor_version TEXT NULL
resolver_version TEXT NULL
legal_status TEXT NULL
allowed_use_classification TEXT NOT NULL
retention_expires_at TIMESTAMP NULL
synth_label TEXT NULL               -- SYNTH_ESTABLISHED | SYNTH_HYPOTHESIS |
SYNTH_UNRESOLVED | SYNTH_GAP | SYNTH_CLOSED | SYNTH_CONFLICT
confidence_band TEXT NULL
CHECK (synth_label IS NULL OR confidence_band IS NULL)
CHECK (synth_label IS NULL OR source_family = 'SYNTHESIS_ASSESSMENT_SOURCE')
```

16.4 Review actions

```
CONFIRM_SIGNAL_USEFUL
DENY_SIGNAL
MARK_ASSOCIATION_SIGNAL          # renamed from MARK_ASSOCIATION_ONLY
REQUEST_MORE_EVIDENCE
ESCALATE_TO_EDD
ATTACH_SUPPORTING_EVIDENCE
SUPPRESS_SIMILAR_SIGNAL
```

17. Inference, contradiction and temporal lineage pipeline

17.1 Candidate generation

- Generate candidates only after entity resolution reaches the required threshold (or the binding is inherited from synthesis).
- Use explicit typed patterns: role history, formal/claimed ownership, nominee language, de facto control, related-party finance, asset control and proxy operation.
- Keep association candidates separate from ownership/control candidates.
- Do not generate a candidate for a proposition that is `SYNTH_ESTABLISHED` on the same binding; reference the Assessment instead.
- Prefer candidates that address a `SYNTH_HYPOTHESIS` indicator.

17.2 Corroboration independence

IOI must identify syndicated or copied claims. Ten articles repeating one investigation are one origin, not ten independent corroborations. Evidence lineage retains origin clustering and citation ancestry where detectable. A synthesis Assessment counts as **one** S1 item per cited assertion, never as multiple corroborations because it cites multiple facts.

17.3 Contradiction handling

Contradiction type	Example	Effect
Synthesis closure	Synthesis `CLOSED(FALSE)` on the same proposition/binding	**Mandatory** `CONTRADICTED`; synthesis assertion listed first
Synthesis conflict	Synthesis `CONFLICT` between admissible facts	Surface as `SYNTH_CONFLICT`; IOI does not resolve
Identity contradiction	Two people with similar names are conflated	Reject or cap until resolved
Temporal contradiction	Profile says role ended before alleged control period	Cap or mark stale
Source contradiction	Registry/company filing denies or supersedes media claim	Mark CONTRADICTED and surface both
Legal-status contradiction	Allegation is later dismissed or overturned	Update status; retain history
Relationship contradiction	Evidence supports employment only, not control	Downgrade to ASSOCIATION_SIGNAL

17.4 Temporal lineage

All role and control signals are time-bounded where dates are available. IOI must be able to state that a person was associated with an entity during a defined period and that current ownership remains unknown. Old evidence must not silently imply present control. Synthesis-derived tenures carry the Assessment's `RESIGNED` / `CENSORED` end kinds and `observed_through`; IOI does not re-derive them.

18. Nyansapo API contracts

18.1 Entity-level signals

`GET /v1/entities/{golden_id}/ioi-signals`

```
        {
          "entity_id": "golden_123",
          "module": "IOI",
          "decision_effect": "CONTEXT_ONLY",
          "confirmed_ownership": false,
          "synthesis_basis": {
            "assessment_id": "asm-2026-1013-000042",
            "execution_id": "…",
            "snapshot_id": "…",
            "as_of": "2026-10-13T09:00Z",
            "status": "CONSUMED"
          },
          "synthesis_references": [
            {"assertion_key": "…", "label": "SYNTH_ESTABLISHED", "predicate":
"SHARED_OFFICER_ESTABLISHED", "subjects": ["person_456", "company_123"]}
          ],
          "signals": [
            {
              "signal_id": "ioi_sig_001",
              "relationship_type": "INFERRED_CONTROLLER",
              "from": {"golden_id": "person_456", "display_name": "Person A"},
              "to": {"golden_id": "company_123", "display_name": "Company B"},
              "confidence_band": "MEDIUM",
              "review_status": "PENDING",
              "source_basis": ["COURT_RECORD_SOURCE", "COMPANY_WEB_SOURCE",
"MEDIA_SOURCE"],
              "corroboration_count": 2,
              "contradiction_count": 1,
              "addresses_indicator": {"assertion_key": "…", "precondition_id": "P3"},
              "constrained_by": [],
              "summary": "Sources suggest possible control. Ownership is not confirmed and
control is not established by Synthesis.",
              "required_copy": "Review-only. Does not establish confirmed ownership."
            }
          ]
        }
```


18.2 Transaction-bundle evaluation

`POST /v1/ioi/transaction-bundles/evaluate` — request contains only the governed cross-document contract, resolved parties, trigger types and evidence pointers. Raw documents remain in the Veredis scan pipeline. IOI performs step 0 per resolved `golden_id`.

18.3 Review event

`POST /v1/ioi/signals/{signal_id}/reviews`


```
        {
          "action": "MARK_ASSOCIATION_SIGNAL",
          "reason": "Known logistics relationship; no ownership connection",
          "veredis_scan_id": "scan_001",
          "reviewer_tenant_id": "tenant_123"
        }
```

18.4 Admin trace

`GET /admin/v1/ioi/signals/{signal_id}/trace` — includes the consumed `assessment_id`, the `SYNTH_*` items, the indicator addressed, and a link to the synthesis explain payload for each referenced assertion.

18.5 Ingest candidate (internal)

`POST /admin/v1/ioi/ingest-candidates` — IOI → governed source admission; body `{candidate_type: "POSSIBLE_SYNTHESIS_EVIDENCE", source_family, source_name, evidence_item_id, rationale}`.

19. Veredis consumption and orchestration contract

19.1 Veredis owns the experience, not the inference

- Determine whether scan policy requires an IOI request.
- Supply resolved entity IDs and governed cross-document context.
- Render the returned IOI projection without changing signal semantics; render `synthesis_references` as established facts, visually and textually distinct from IOI signals.
- Expose evidence, contradictions, the indicator addressed, and mandatory caveats.
- Capture user actions and submit them to Nyansapo.
- Preserve deterministic decision independence: neither IOI nor Synthesis can override or alter the scan verdict.

19.2 Trigger policy

Trigger	Request IOI?
AOT returns a complete confirmed path	Optional context only
AOT returns no path or incomplete upstream ownership	Yes
Synthesis returns `HYPOTHESIS` / `UNRESOLVED` on a material proposition	Yes (indicator-targeted)
Synthesis returns `EMITTED` on the material proposition	Optional; IOI will reference, not compete
Cross-doc finds supplier/payee or seller/shipper mismatch	Yes when policy threshold is met
Repeated unknown intermediary across bundle documents	Yes
RRI returns a material control-adjacent relationship	Yes
Controlled-goods or sanctions context creates enhanced review need	Yes, context-only
No ownership gap and no relationship trigger	No or tenant-policy controlled

19.3 Consumption flow

```
Veredis document/bundle intake
  -> cross-document reconciliation
  -> entity resolution through Nyansapo
  -> AOT request
  -> (Synthesis request, if Veredis consumes it directly for its own Bounded Intelligence
Summary)
  -> Veredis IOI trigger policy
  -> Nyansapo IOI evaluation (IOI performs its own step-0 synthesis consumption)
  -> Veredis Web/PDF/API rendering
  -> user review action
  -> Nyansapo IOI review event
```

20. Veredis output contracts

20.1 Dashboard

A separate review-only IOI module: signals list, highest band, contradictions, review status; a distinct **Established by Synthesis** block above the IOI signals listing `synthesis_references` with their predicates and the Assessment id; cross-document triggers; review controls; mandatory disclaimer. The two blocks must not be visually merged.

20.2 Entity result — text contract

```
ESTABLISHED (Synthesis asm-2026-1013-000042, as of 2026-10-13)
  SHARED_OFFICER_ESTABLISHED: Person A — Company B (appointments overlap 412 days)

IOI — Inferred Ownership Signals                                REVIEW-ONLY

Possible control signal
Person A -> Company B
Confidence: Medium
Addresses synthesis indicator: P3 (control evidence not present in admissible structured
sources)
Evidence: Court record, company website, investigative media
Contradictions: 1
Status: Pending review

Sources suggest Person A may exercise informal control or influence over Company B.
This has not been confirmed through AOT and has not been established by Synthesis.

[Confirm useful signal] [Deny] [Association signal] [Request more evidence] [Escalate to
EDD]
```

20.3 Cross-document result — text contract

```
Cross-document trigger: Supplier differs from payment beneficiary
Supplier A <-> Payee C
Synthesis: no established relationship (UNRESOLVED — address evidence absent in release R)
IOI signal: Possible payment proxy relationship
Confidence: Low
Source basis: Company website and historical media
Decision effect: CONTEXT_ONLY

Transaction-bundle co-occurrence is not ownership evidence.
This signal requires user review.
```

20.4 PDF dossier block

```
Inferred Ownership Signals

This section contains review-only inferred ownership/control signals produced by Nyansapo
IOI.
Established facts from Nyansapo Synthesis are listed separately above and are not IOI
signals.
The signals do not establish confirmed ownership and do not alter the deterministic
Veredis decision.

Signal IOI-001
Relationship type: Inferred Controller
Confidence: Medium
Synthesis basis: asm-2026-1013-000042 (addresses indicator P3)
Evidence basis: Court record; company leadership page; media report
Contradictions: 1
Decision effect: Context only
Recommended action: Review supporting evidence before reliance.
```

20.5 Structured output

```
        {
          "synthesis": {
            "assessment_id": "asm-2026-1013-000042",
            "established": ["<assertion_key>"],
            "decision_effect": "CONTEXT_ONLY"
          },
          "ioi": {
            "status": "SIGNALS_FOUND",
            "decision_effect": "CONTEXT_ONLY",
            "confirmed_ownership": false,
            "established_by_synthesis": false,
            "review_required": true,
            "signals": ["ioi_sig_001"]
          }
        }
```

20.6 Copy guard

Build-time and request-time enforcement of §5.1, §5.2 and §5.5: forbidden phrases fail the build; cross-vocabulary misuse (an IOI signal described as *established*; a `SYNTH_ESTABLISHED` item described as *possible*; a band on a `SYNTH_*` item) fails the request. Web, PDF and structured outputs must pass the same guard with identical results (parity test §25.3).

21. User review and feedback loop

The review loop improves usefulness, ranking and suppression patterns. It does not convert a signal into confirmed ownership, does not create or alter a synthesis fact, and does not permit deterministic decision effects.

Action	Meaning	State effect
CONFIRM_SIGNAL_USEFUL	Reviewer considers the signal relevant for the case	Records usefulness; remains IOI
DENY_SIGNAL	Reviewer finds the inference incorrect	Records rejection and reason
MARK_ASSOCIATION_SIGNAL	Relationship exists but does not support control/ownership	Downgrades relationship class
REQUEST_MORE_EVIDENCE	Evidence is insufficient	Queues retrieval or analyst work
ESCALATE_TO_EDD	Reviewer opens enhanced due diligence	Case workflow only; not IOI decision authority
ATTACH_SUPPORTING_EVIDENCE	Reviewer supplies additional material	New evidence enters validation pipeline
SUPPRESS_SIMILAR_SIGNAL	Reviewer requests suppression of a pattern or source combination	Policy-scoped; audited

> **Hard distinction:** A user-confirmed IOI signal is neither AOT-confirmed ownership nor a synthesis-established fact. Only separately admitted evidence and AOT governance can create a confirmed ownership edge; only synthesis rules over admissible structured facts can establish a proposition.

22. Security, privacy, retention and access controls

22.1 Data minimisation

- Collect business-role and relationship evidence required for IOI; exclude irrelevant private-life data.
- Redact personal contact details unless independently necessary and policy-approved.
- Store source URLs in hashed/redacted form where appropriate while preserving retrievability under controlled access.
- Keep raw Veredis transaction documents outside IOI persistence unless a separate evidence-admission contract applies.
- Synthesis Assessments are stored by reference (`assessment_id`, `assertion_key`, `canonical_sha256`); IOI does not copy full Assessment bodies.

22.2 Access control

- Nyansapo service roles separate source ingestion, IOI read, IOI review, admin trace, and synthesis consumption (`INTEL_READ` , internal-service audience).
- Veredis receives only tenant-safe projected evidence and cannot query unrestricted source corpora.
- Analyst/admin evidence access is audited.
- Professional-profile and sensitive-source adapters are individually feature-flagged.
- IOI has no write grant on any `intel_*` table (CI zero-grant check).

22.3 Retention and decay

- Each source family defines default retention and revalidation intervals.
- Role and control signals expire, decay or become STALE when unsupported by current evidence; `SYNTH_SUPERSEDED` triggers re-evaluation.
- Review events are retained as immutable audit history.
- Erasure, suppression and correction workflows must not destroy the audit reason for a decision unless legally required.

23. Auditability, observability and operations

23.1 Audit trace

```
        {
          "module": "IOI",
          "signal_id": "ioi_sig_001",
          "decision_effect": "CONTEXT_ONLY",
          "confirmed_ownership": false,
          "established_by_synthesis": false,
          "synthesis_basis": {"assessment_id": "asm-…", "execution_id": "…", "status":
"CONSUMED"},
          "sources_evaluated": 12,
          "sources_admitted": 4,
          "synth_items": {"established": 2, "hypothesis": 1, "closed": 0, "conflict": 0},
          "model_versions": {"extractor": "ioi-rel-v1", "resolver": "resolver-current",
"ranker": "ioi-ranker-v1"},
          "guards": {"aot_write_blocked": true, "synth_write_blocked": true,
"decision_cap_enforced": true, "review_only_copy_enforced": true,
"vocabulary_guard_enforced": true}
        }
```

23.2 Operational metrics

Metric	Purpose
ioi_candidates_generated_total	Candidate volume by source family and relationship type
ioi_signals_admitted_total	Signals surviving evidence and policy gates
ioi_signals_contradicted_total	Contradiction rate and source-quality indicator
ioi_signals_synth_constrained_total	Signals forced `CONTRADICTED` by a synthesis closure

Metric	Purpose
ioi_signals_addressing_indicator_total	Signals that target a synthesis indicator
ioi_synth_unavailable_total	Step-0 failures; degraded-mode evaluations
ioi_synth_superseded_total	Signals re-queued after a newer Assessment
ioi_ingest_candidates_filed_total	Reverse-channel volume
ioi_entity_resolution_reject_total	Identity quality and false-link prevention
ioi_source_policy_quarantine_total	Unapproved or review-required acquisitions
ioi_review_outcomes_total	User/admin review outcome distribution
ioi_aot_write_attempt_block_total	Critical isolation guard
ioi_synth_write_attempt_block_total	Critical isolation guard
ioi_decision_cap_violation_total	Critical decision-effect guard
ioi_vocabulary_guard_fail_total	Cross-vocabulary misuse caught
ioi_projection_parity_fail_total	Web/PDF/API contract divergence

23.3 Failure posture

- IOI unavailable: Veredis continues without IOI and states that inferred intelligence is unavailable; it does not fabricate a result.
- Synthesis unavailable: IOI runs in degraded mode (§10.1) and says so; no `SYNTH_*` constraints applied; signals flagged `synthesis_status = UNAVAILABLE`.
- Ceres unavailable: IOI may use other sources, but legal-source-dependent signals remain incomplete.
- Cross-doc contract unavailable: entity-level IOI may operate; bundle-triggered IOI is not called.
- Guard failure: fail closed for IOI output and alert operations.

24. Feature flags, sequencing and rollout

24.1 Nyansapo flags

```
NYA_IOI_ENABLED=false
NYA_IOI_SYNTHESIS_CONSUMPTION_ENABLED=false
NYA_IOI_SYNTHESIS_REQUIRED=true           # when true, step-0 failure = degraded mode with
explicit status; never silent
NYA_IOI_SOURCE_ACQUISITION_ENABLED=false
NYA_IOI_CERES_GROUNDING_ENABLED=false
NYA_IOI_RRI_FEATURE_INPUT_ENABLED=false
NYA_IOI_ENTITY_SIGNALS_ENABLED=false
NYA_IOI_TRANSACTION_BUNDLE_ENABLED=false
NYA_IOI_PROFESSIONAL_PROFILE_SOURCE_ENABLED=false
NYA_IOI_REVIEW_LOOP_ENABLED=false
NYA_IOI_INGEST_CANDIDATES_ENABLED=false
NYA_IOI_PUBLIC_API_ENABLED=false
NYA_IOI_MAX_DECISION_EFFECT=CONTEXT_ONLY
```

24.2 Veredis flags

```
VER_IOI_CONSUMPTION_ENABLED=false
VER_IOI_TRIGGER_POLICY_ENABLED=false
VER_IOI_WEB_RENDER_ENABLED=false
VER_IOI_PDF_RENDER_ENABLED=false
VER_IOI_STRUCTURED_OUTPUT_ENABLED=false
VER_IOI_REVIEW_ACTIONS_ENABLED=false
VER_IOI_CROSS_DOC_BUNDLE_ENABLED=false
VER_IOI_SYNTH_REFERENCE_RENDER_ENABLED=false
```

24.3 Sequencing

Phase	Scope	Dependency/status
0 — Boundary and contract seal	Architecture, vocabulary separation, copy rules, source registry, schemas, API contracts	Approved future scope; design may proceed
1 — Nyansapo core	Isolated schema, guards (incl. synthesis isolation and vocabulary guard), evidence lineage, admin trace	After current higher-severity committed work
2 — Synthesis adapter	Step-0 consumption, `SYNTH_*` mapping, constraints, supersession check, degraded mode	**After NYA-INTEL-SYNTH-001 first slice is accepted**
3 — Source prototype	Court, insolvency, registry, company web and media adapters; indicator-targeted jobs	Source-specific approval
4 — Ceres adapter	Grounded legal passages into IOI evidence contract	Ceres contract stable
5 — RRI feature adapter	Consume approved relationship features	RRI contract stable
6 — Cross-doc adapter	Consume governed bundle entities and trigger patterns	Cross-document contract stable
7 — Ingest-candidate channel	Reverse channel to governed source admission	Source admission process ready
8 — Veredis internal UAT	Web/PDF/API rendering incl. synthesis-reference block, review loop	Nyansapo APIs and copy guard sealed
9 — External UAT	Selected users; review-only controls and monitoring	Internal UAT accepted
10 — Controlled release	Specific tenants/use cases under feature flags	Governance approval

25. Acceptance criteria and negative tests

25.1 Architecture acceptance

Test	Expected result
IOI module placement	Independent Nyansapo bounded context; not under Ceres, Synthesis or Veredis
Ceres response	Passages and source status only; no inferred relationship verdict
Veredis role	Requests, renders and submits review; no edge generation
Review persistence	Authoritative event stored in Nyansapo
AOT write attempt	Blocked and audited
Synthesis write attempt (any `intel_*` table)	Blocked and audited
IOI decision above CONTEXT_ONLY	Blocked and audited
Signal type name equals a synthesis predicate	CI fails

25.2 Evidence and inference tests

Scenario	Expected result
Synthesis `EMITTED` shared-officer on (A,B); IOI has media claiming the same	No IOI signal for that proposition; `synthesis_references` lists the assertion; media item stored as corroborated-by-synthesis
Synthesis `CLOSED(FALSE)` on (A,B); IOI has two S3 sources asserting it	IOI signal emitted with `CONTRADICTED`; synthesis assertion first in contradictions; never suppressed silently
Synthesis `HYPOTHESIS` with indicator P3; IOI finds a court passage addressing P3	Signal carries `addresses_indicator = P3`; copy states synthesis could not establish because P3 evidence was absent
Synthesis unavailable	Degraded mode; `synthesis_status = UNAVAILABLE`; no fabricated basis; rendering states it
Newer Assessment supersedes the consumed one	Signal → `SYNTH_SUPERSEDED`; re-evaluation queued
`SYNTH_*` item given a confidence band	DB check constraint rejects
Single weak blog claims X owns Y	Reject or LOW; no HIGH signal
Professional profile says CEO	Executive affiliation only
Professional profile says owner	Review-only self-asserted ownership; confidence capped
Court pleading alleges control without judgment	Legal status = ALLEGATION; review-only
Allegation later dismissed	Contradiction/supersession displayed
Ten syndicated articles repeat one source	One origin cluster, not ten corroborations
One Assessment citing six facts	One S1 item per cited assertion; not six corroborations of each other
Same address only	ASSOCIATION_SIGNAL unless corroborated; synthesis co-location in release R referenced if present

Scenario	Expected result
Cross-doc supplier/payee mismatch only	Trigger evaluation; no inferred edge without external evidence
User confirms signal	Remains IOI; no AOT promotion; no synthesis fact
Old founder profile only	STALE or historical role; no current-control assertion

25.3 Veredis output tests

Test	Expected result
Web output	Separate “Established by Synthesis” block and separate IOI section with mandatory review-only caveat
PDF output	Same structure, labels, confidence, evidence and caveat as Web
Structured output	`decision_effect=CONTEXT_ONLY`; `confirmed_ownership=false`; `established_by_synthesis=false` on every IOI signal
Contradiction present	Prominently rendered; synthesis closure first
IOI unavailable / Synthesis unavailable	No fabricated signals or basis; availability state shown
Forbidden phrase or cross-vocabulary misuse generated	Copy/vocabulary guard fails build/request
Combined score across synthesis and IOI	Not computed; parity test asserts absence

26. Implementation ticket breakdown

26.1 Nyansapo

```
NYA-IOI-001              IOI bounded-context architecture and ADR (v1.2)
NYA-IOI-VOCAB-001        Vocabulary separation, SYNTH_* labels, rename ASSOCIATION_ONLY →
ASSOCIATION_SIGNAL, CI predicate-collision check
NYA-IOI-SCHEMA-001       Isolated schema incl. synthesis_basis, synth_label constraints,
consumption/jobs/ingest-candidate tables
NYA-IOI-GUARDS-001       AOT isolation, synthesis isolation (zero intel_* grants),
decision-effect cap, vocabulary guard
NYA-IOI-SYNTH-001        Synthesis adapter: step-0 request, SYNTH_* mapping, constraints,
degraded mode, supersession check
NYA-IOI-SOURCE-REG-001   Source family registry and admission policy (incl.
SYNTHESIS_ASSESSMENT_SOURCE)
NYA-IOI-ACQ-001          Governed source acquisition adapters; indicator-targeted jobs
NYA-IOI-EVIDENCE-001     Evidence lineage and origin clustering
NYA-IOI-RESOLVE-001      Person/company entity linkage; inherited SYNTH bindings
NYA-IOI-TEMPORAL-001     Executive role and control timeline; synthesis tenure passthrough
NYA-IOI-EXTRACT-001      Relationship extraction and typed candidates (excluding
SYNTH_ESTABLISHED propositions)
NYA-IOI-SCORE-001        Confidence, corroboration and contradiction (SYNTH_CLOSED
mandatory CONTRADICTED)
NYA-IOI-CERES-001        Ceres evidence-grounding adapter
NYA-IOI-RRI-001          RRI graph-feature adapter
NYA-IOI-XDOC-001         Cross-document governed-context adapter
NYA-IOI-INGEST-001       Ingest-candidate reverse channel to governed source admission
NYA-IOI-REVIEW-001       Review events and pattern feedback
NYA-IOI-API-001          Entity, bundle, review, admin and ingest-candidate endpoints
NYA-IOI-AUDIT-001        Audit trace and operational metrics
```

26.2 Veredis

```
VER-IOI-001              Nyansapo IOI client and consumption contract
VER-IOI-TRIGGER-001      Entity and bundle IOI orchestration policy (incl. synthesis-
status triggers)
VER-IOI-WEB-001          Web output: established-by-synthesis block + IOI section +
evidence review UI
VER-IOI-PDF-001          PDF dossier rendering parity
VER-IOI-API-001          Structured/VZP output
VER-IOI-XDOC-001         Cross-document IOI presentation
VER-IOI-REVIEW-001       User review actions and feedback submission
VER-IOI-COPY-GUARD-001   Forbidden wording, vocabulary separation and caveat enforcement
VER-IOI-QA-001           QA/UAT test pack and parity gates (incl. no-combined-score
assertion)
```

26.3 Shared contracts

```
NYA-CERES-IOI-GROUND-001   Passage grounding contract for IOI
NYA-GRAPH-IOI-TYPES-001    Typed inferred-edge projection contract
NYA-XDOC-IOI-HANDOFF-001   Stable bundle-entity and trigger handoff contract
NYA-SYNTH-IOI-CONSUME-001  Synthesis consumption contract (accommodations requested from
the synthesis programme are listed in the accompanying dev-team message; none change
synthesis vocabulary)
```

27. Founder decisions required for v1.2

ID	Decision	Recommendation
FD-IOI-1	Adopt v1.2 as the complete replacement for v1.1 (no reader needs v1.1).	Yes.
FD-IOI-2	Synthesis as mandatory step 0, with degraded mode (never silent) when unavailable.	Yes; `NYA_IOI_SYNTHESIS_REQUIRED=true`.
FD-IOI-3	Vocabulary separation enforced on the IOI side only; synthesis vocabulary untouched. Rename `ASSOCIATION_ONLY` → `ASSOCIATION_SIGNAL`; `SYNTH_*` labels; CI predicate-collision check.	Yes.
FD-IOI-4	Synthesis `CLOSED(FALSE)` forces `CONTRADICTED`; synthesis `EMITTED` suppresses a competing IOI signal in favour of a reference.	Yes.
FD-IOI-5	Reverse channel limited to ingest candidates through governed source admission.	Yes.
FD-IOI-6	IOI sequenced after the synthesis first slice (in addition to RRI and cross-doc).	Yes; dependency recorded in §24.3 phase 2.
FD-IOI-7	Veredis must not compute any combined score across synthesis and IOI.	Yes; parity test.

Appendix A. Candidate source inventory

Source class	Typical content	IOI treatment
Synthesis Assessments	Established/closed/conflicted propositions; hypotheses with indicators	S1; unbanded; references and constraints
Company registries	Officers, directors, beneficial-owner filings, charges, status and history	AOT candidate or IOI support; cite synthesis fact where the same release is already consumed

Source class	Typical content	IOI treatment
GLEIF/LEI	Declared parent and relationship records	Entity lineage and corroboration
Securities filings	Issuer reports, holdings, related parties and subsidiaries	High-value ownership and control evidence
Court records	Judgments, pleadings, asset tracing and de facto control language	Legal-status-aware IOI evidence; ingest-candidate source
Insolvency/bankruptcy	Administrator reports, creditor structures and related-party transactions	Hidden interest and control signals; ingest-candidate source
Regulatory enforcement	Enforcement findings, sanctions evasion and asset freezes	High-value network/control evidence
Procurement	Awards, contractors and recurring supplier relationships	Operating-network signals
Asset registries	Vessels, aircraft, land, concessions and licences	Asset-control signals
Company websites	Leadership, founder and board pages	Official self-published role lineage
Media and trade press	Investigations, interviews and historic company narratives	Corroborated secondary evidence
NGO reports	Anti-corruption and sanctions-evasion investigations	Network/control allegations
Professional profiles	Role chronology, founder and board claims	Self-asserted, confidence-capped
Archived web	Removed biographies and historical company pages	Temporal continuity and succession
Business directories	Public executive/company listings	Weak support or discovery
Search indexes	Discovery of underlying sources	Never evidence by itself

Appendix B. Payload and output examples

B.1 Cross-document request

```
        {
          "bundle_id": "txn_bundle_2026_001",
          "reconciled_entities": [
            {"bundle_entity_id": "be_supplier", "role": "supplier", "golden_id":
"golden_123"},
            {"bundle_entity_id": "be_payee", "role": "payment_beneficiary", "golden_id":
null}
          ],
          "cross_doc_relationships": [
            {"relationship_type": "SUPPLIER_DIFFERS_FROM_PAYEE", "from": "be_supplier",
"to": "be_payee"}
          ],
          "audit_trace_id": "xdoc_trace_001"
        }
```

B.2 IOI response


```
        {
          "bundle_id": "txn_bundle_2026_001",
          "module": "IOI",
          "decision_effect": "CONTEXT_ONLY",
          "synthesis_basis": {"golden_123": {"assessment_id": "asm-…", "status":
"CONSUMED"}, "be_payee": null},
          "signals": [
            {
              "signal_id": "ioi_sig_txn_001",
              "relationship_type": "INFERRED_PAYMENT_PROXY",
              "confidence_band": "LOW",
              "cross_doc_trigger": "SUPPLIER_DIFFERS_FROM_PAYEE",
              "summary": "External evidence suggests a possible operating relationship.
Ownership is not confirmed and no relationship is established by Synthesis.",
              "review_status": "PENDING"
            }
          ]
        }
```


B.3 No-signal response


```
        {
          "status": "NO_REVIEWABLE_SIGNALS",
          "decision_effect": "CONTEXT_ONLY",
          "confirmed_ownership": false,
          "synthesis_references": ["<assertion_key>"],
          "required_copy": "No reviewable inferred ownership signals were identified from
admitted sources. Established facts, if any, are listed under Synthesis."
        }
```


Appendix C. Architecture decision record

ADR field	Decision
Title	Place IOI as an independent Nyansapo bounded context that consumes Synthesis first
Status	Accepted for future implementation; sequenced after RRI, cross-doc and the synthesis first slice; not deferred
Context	IOI requires source lineage, canonical identity, temporal graph state, contradiction handling, review persistence and a deterministic baseline of what is already established or disproved. Ceres is too narrow; Veredis is too downstream; Synthesis is deterministic and must not absorb banded confidence or weak sources.

ADR field	Decision
Decision	Nyansapo owns IOI. Synthesis supplies the signed established/closed baseline and the indicators IOI targets. Ceres supplies grounded passages. Veredis owns request policy, rendering and review submission. Vocabulary separation is enforced on the IOI side.
Consequences	No split-brain graph; no second “inference” meaning; stronger AOT and synthesis isolation; additional adapter, schema and guard work; IOI cannot ship before the synthesis slice.
Rejected option 1	Implement IOI inside Ceres — Ceres is retrieval/grounding, not probabilistic graph-state ownership.
Rejected option 2	Implement IOI in Veredis — duplicates substrate responsibilities and sits next to deterministic decisions.
Rejected option 3	Implement IOI inside Synthesis as a rule family with confidence — would admit S3/S4 sources, probability and ranking models into the deterministic engine and break its constitutional prohibitions.

> **Final design position:** IOI is a Nyansapo intelligence engine adjacent to AOT, Synthesis, RRI and Ceres. Synthesis establishes; IOI infers only where synthesis is silent or unresolved, is constrained by what synthesis has closed, and never feeds synthesis anything but ingest candidates. Ceres grounds. Veredis orchestrates and renders. IOI remains review-only, source-backed and isolated from confirmed ownership, established facts and deterministic decision effects.

---

v1.2 generated 13 October 2026. Complete document; supersedes v1.1. Confidential working specification — IOI is review-only and CONTEXT_ONLY.